

Assignment

Title: Blockchain-Based Smart Agricultural Supply Chain

Introduction

Agriculture is one of the most important sectors of the economy, and ensuring the quality, safety, and authenticity of agricultural products is essential. Traditional agricultural supply chains involve multiple stakeholders such as farmers, warehouses, transporters, retailers, and consumers. Since these processes are often managed using paper-based records or centralized databases, they may suffer from problems such as lack of transparency, data manipulation, delays in product tracking, and food fraud.

Blockchain technology provides a decentralized, secure, and tamper-proof platform for recording and sharing information among all participants in the supply chain. Every transaction is stored as an immutable block, making it easy to trace the complete journey of a product from the farmer to the consumer. Smart contracts automate important business processes, while cryptographic techniques ensure data security and integrity.

Problem Statement

Design a blockchain-based system to track agricultural products from farmers to consumers. The system should improve transparency, product quality, and trust among all stakeholders. Traditional agricultural supply chains rely heavily on paper-based records and multiple intermediaries, which creates opportunities for data manipulation, delays in traceability, and a general lack of trust between the parties involved — particularly the end consumer, who has almost no visibility into how a product reached them.

Real-Time Scenario

A farmer produces rice. The rice is transported through distributors and retailers before reaching the customer. Every stage of the journey should be recorded on the blockchain. For example, when the farmer harvests the rice, the batch details are logged on-chain. When the batch moves to a warehouse for storage and then to a transporter for delivery, GPS location and environmental conditions are recorded automatically via IoT sensors. Finally, when the rice reaches the retail store and is purchased, the consumer can scan a QR code on the packaging to view the complete, verified history of that specific batch — including where it was grown, how it was stored, and how it was transported — before deciding to buy it.

Objectives

- To analyze existing agricultural supply chain challenges related to transparency and traceability.
- To design a Blockchain-based framework for tracking products from farmers to consumers.
- To identify appropriate Blockchain architectures and consensus mechanisms for supply chain applications.
- To evaluate how Blockchain improves product authenticity and operational efficiency.

Stakeholder Identification and Roles

The table below lists all participants involved in the blockchain-based agricultural supply chain and explains their respective roles and responsibilities.

Stakeholder	Role / Responsibility
Farmer	Produces the rice crop; records details such as crop type, sowing/harvest date, quantity, and initial quality grading onto the blockchain at the point of origin.
Warehouse	Receives produce from farmers, verifies quantity and quality, records storage conditions (temperature, humidity), and assigns a batch ID that is logged on-chain.
Transporter	Moves the produce from warehouse to retailer; logs GPS location, route, transit time, and environmental conditions (temperature/humidity) using IoT sensors linked to the blockchain.
Retailer	Receives goods from the transporter, verifies the batch against blockchain records, updates shelf/stock information, and sets the final retail price.
Consumer	Scans a QR code at the point of sale to view the complete, tamper-proof history of the product — from farm to shelf — before making a purchase decision.
Quality / Certification Authority	Independently verifies quality and safety standards at key checkpoints (e.g., organic certification, pesticide testing) and records certification results on the blockchain.
Blockchain Network (Consortium Nodes)	Validates and stores every transaction immutably, contracts, and enforces consensus among all participating organizations.



Fig. 1 — Stakeholder Connectivity via Consortium Blockchain

Proposed Blockchain System Architecture

The proposed architecture is organized into four layers. The Application Layer provides role-specific portals/apps for each stakeholder. The Data Capture Layer uses IoT sensors (GPS, temperature, humidity) and QR/barcode scanning to collect real-world data automatically, minimizing manual entry errors. This data is submitted to the Consortium Blockchain Network, where smart contracts validate and record every transaction as an immutable block. Finally, the Storage & Services Layer keeps large files (such as certificates and images) off-chain via IPFS/cloud storage while storing only their cryptographic hashes on-chain, alongside identity management and an analytics dashboard for reporting.

The proposed architecture ensures that every stakeholder has access to accurate, secure, and real-time information throughout the agricultural supply chain. By integrating IoT devices with blockchain technology, the system enables continuous monitoring of product quality, storage conditions, and transportation status while maintaining complete data integrity. Smart contracts automate business processes such as product registration, quality verification, and ownership transfer, reducing manual intervention and operational delays. and trust among all participating organizations.

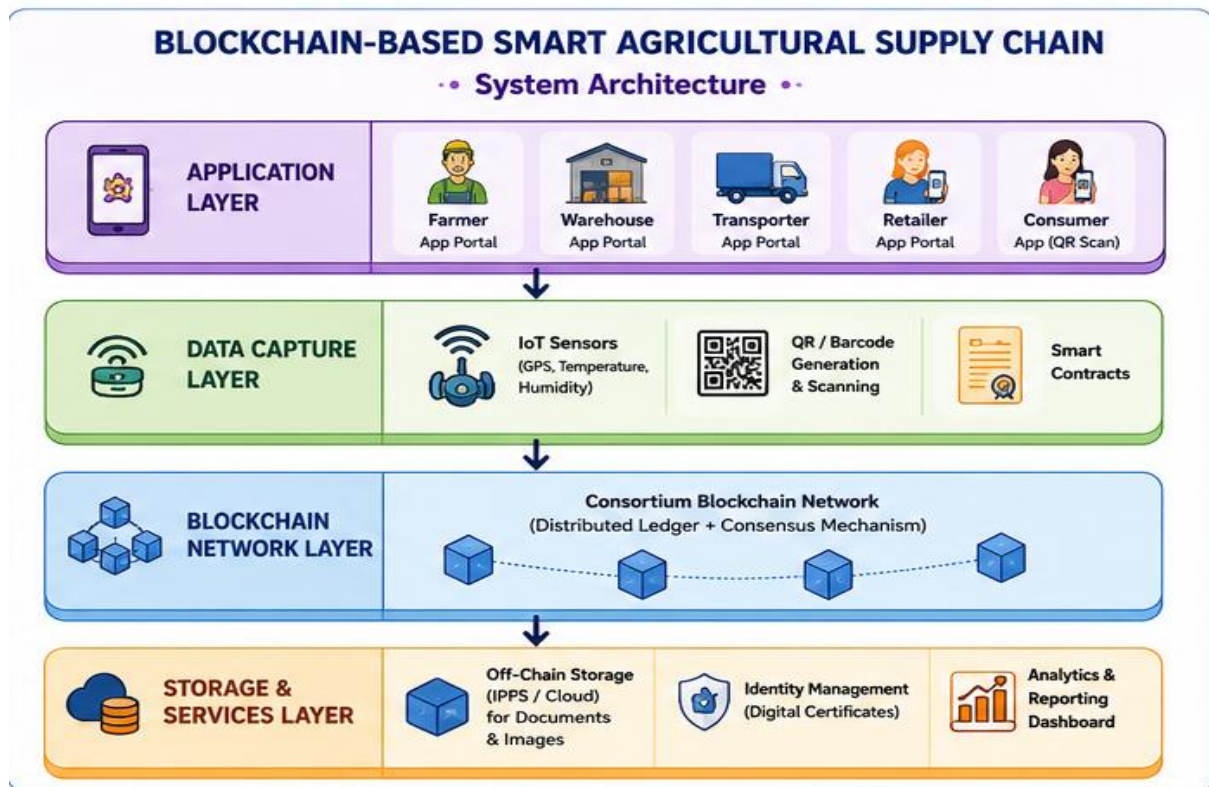


Fig. 2 — System Architecture Diagram

Workflow Diagram

The diagram below illustrates the journey of rice from the farmer to the customer, along with the specific information that is captured and recorded on the blockchain at every stage of the process.



Fig. 3 — Product Flow: Farmer → Warehouse → Transport → Retail Store → Customer

- **Farmer:** Crop type, harvest date, quantity harvested, and initial quality grade are recorded, generating a unique batch ID.
- **Warehouse:** Storage conditions (temperature, humidity), duration of storage, and a post-storage quality check are logged against the batch ID.
- **Transport:** GPS route, dispatch and arrival timestamps, and in-transit environmental conditions are recorded automatically via IoT sensors.
- **Retail Store:** Date of receipt, shelf placement date, retail price, and a final quality verification are recorded.
- **Customer:** Scans the QR code on the package to instantly view the complete, tamper-proof history of the product before purchase.

Blockchain Benefits

Five key benefits of applying blockchain technology to the agricultural supply chain are summarized below.

#	Benefit	Explanation
1	Product Traceability	Every stage of the rice's journey — farm, warehouse, transport, retail — is recorded on an immutable ledger, allowing full end-to-end tracing of any batch.
2	Transparency	All authorized stakeholders can view the same verified data, removing information asymmetry between farmers, distributors, and consumers.
3	Reduced Fraud	Immutable, time-stamped records make it extremely difficult to falsify quality certificates, origin claims, or quantities.
4	Better Food Safety	Contamination or quality issues can be traced back instantly to the exact batch, farm, and stage, enabling faster recalls.
5	Faster Tracking	Automated smart contracts and digital records replace slow manual paperwork, speeding up verification at every checkpoint.

Blockchain Type Selection with Justification

A Consortium Blockchain is the most suitable choice for this agricultural supply chain system, for the following reasons:

- **Controlled Participation:** Only known, verified organizations — farmer cooperatives, warehouses, logistics companies, retailers, and certification bodies — act as validating nodes, which is appropriate since supply chain partners are known business entities rather than anonymous public users.
- **Balanced Transparency and Privacy:** Unlike a fully public blockchain, a consortium model allows sensitive commercial data (e.g., pricing, contract terms) to remain restricted to relevant parties, while product traceability data is still shared with consumers through summarized views.
- **Higher Performance and Lower Cost:** With a limited, trusted set of validator nodes, the network can use lightweight consensus mechanisms (e.g., PBFT or Proof of Authority) that offer faster transaction throughput and negligible fees compared to public blockchains like Ethereum mainnet.
- **Shared Governance:** Since multiple independent organizations (farmer cooperatives, transport companies, retail chains) jointly govern the network, no single entity can unilaterally alter records, which builds trust while avoiding the overhead of a fully public, permissionless system.

Smart Contract Design

The following smart contracts are proposed to automate key stages of the supply chain process and strengthen the overall blockchain solution design.

Smart Contract	Purpose
Produce Registration Contract	Triggered when a farmer registers a new harvest batch; records crop type, quantity, harvest date, and generates a unique batch ID.
Quality Verification Contract	Executes automatically when the Certification Authority uploads a quality/inspection result, updating the batch's verified status onchain.
Custody Transfer Contract	Manages the transfer of custody of a batch between Farmer → Warehouse → Transporter → Retailer, logging timestamps and GPS location at each handover.
Payment Settlement Contract	Automatically releases payment to the farmer or distributor once predefined delivery and quality conditions (e.g., temperature range maintained, on-time delivery) are met.

Security Features

The following security techniques ensure the integrity, authenticity, and confidentiality of data within the agricultural blockchain system.

Security Technique	How it is Applied
Hashing (SHA-256)	Each transaction (harvest record, transport log, quality report) is hashed; any tampering changes the hash and is instantly detectable.
Digital Signatures	Every stakeholder signs their transactions with a private key, proving authenticity and preventing impersonation or repudiation.
Encryption	Sensitive data (e.g., pricing, contractual terms) is encrypted before storage, so only authorized parties with the correct keys can read it.
Access Control (RBAC)	Role-based permissions ensure a transporter can update logistics data but cannot alter quality certifications, and vice versa.
Blockchain Immutability	Once a block is added and confirmed by consensus, records cannot be altered or deleted, preserving a permanent audit trail.
Consensus Mechanism (PBFT/PoA)	Since this is a permissioned consortium network, a lightweight consensus (e.g., Practical Byzantine Fault Tolerance or Proof of Authority) is used among known, trusted organizational nodes to validate transactions efficiently.

Applications

- Rice supply chain management
- Fruit and vegetable distribution
- Dairy product tracking
- Organic food certification
- Coffee and tea supply chains
- Export and import of agricultural products

Challenges

Three key challenges in implementing this blockchain-based agricultural supply chain system are described below.

#	Challenge	Explanation
1	Scalability & Transaction Cost	As the number of farmers, transporters, and retailers grows, the network must handle a high volume of transactions without slowing down or becoming expensive.
2	Digital Literacy & Infrastructure	Many farmers, especially in rural areas, may lack smartphones, internet access, or the technical knowledge to interact with a blockchain-based system.
3	Data Accuracy at Source (Garbage-In Problem)	Blockchain guarantees data is not altered after entry, but it cannot verify that the data entered by a farmer or worker was accurate in the first place.

Conclusion

The proposed Blockchain-Based Smart Agricultural Supply Chain system addresses the core problems of opacity and fraud in traditional agricultural distribution by recording every stage of a product's journey — from the farmer's field to the retailer's shelf — as an immutable, verifiable transaction. By adopting a consortium blockchain model combined with IoT-based automated data capture, smart contracts, and strong cryptographic security, the system significantly improves transparency, traceability, and trust for all stakeholders, including the end consumer. While challenges such as scalability, rural digital literacy, and ensuring data accuracy at the point of entry remain, the overall framework demonstrates how blockchain technology can modernize and secure agricultural supply chains, ultimately benefiting farmers through fair recognition of their produce and consumers through verified food safety and authenticity.